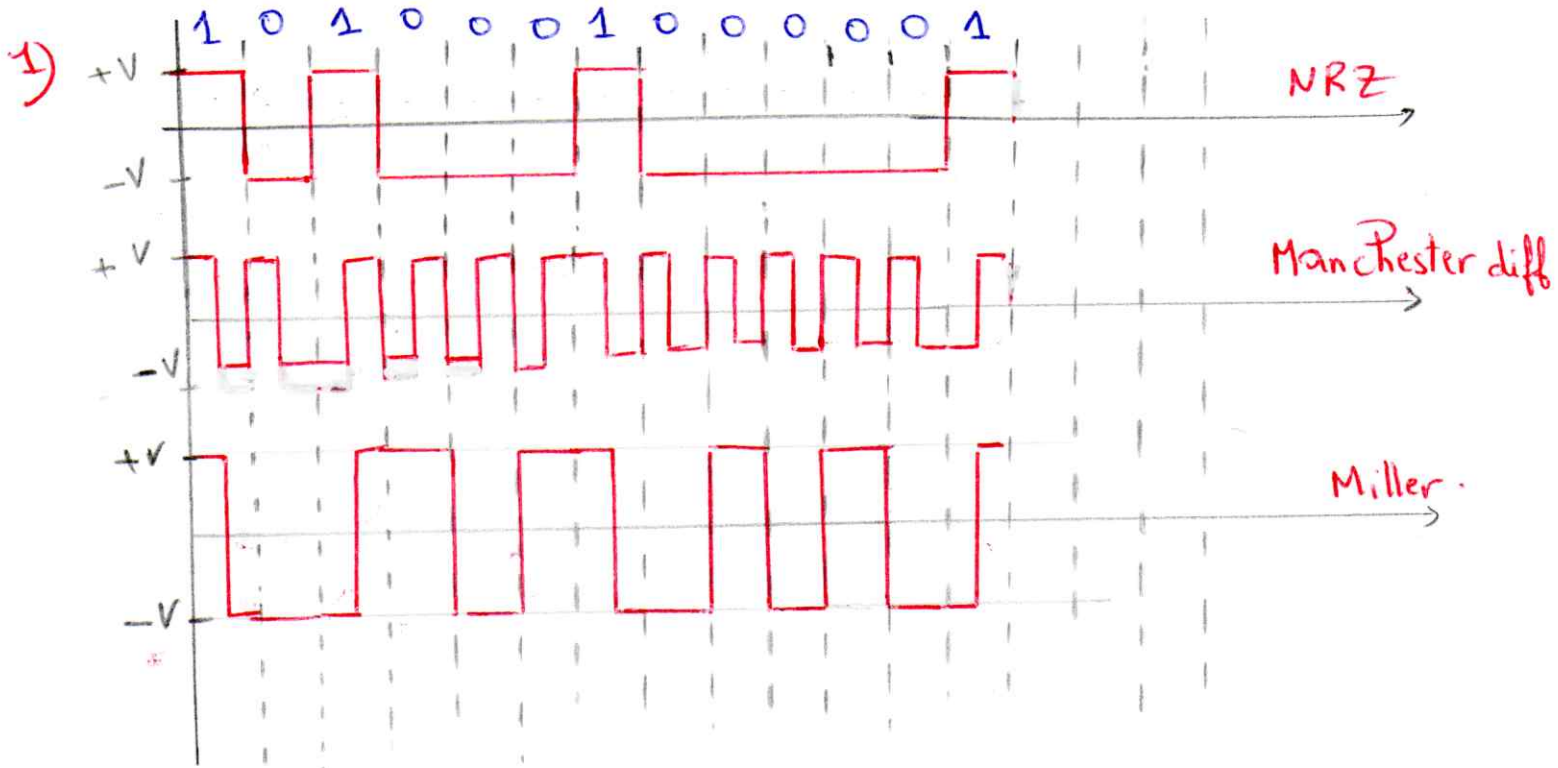


Exercice 1:



2)  $R_m = 8000$  bauds.

\* Cas NRZ:

$$D = \frac{R_m}{k} \log_2(v)$$

or  $v = 2$  et  $k = 1$

$$\Rightarrow D = \frac{R_m}{1} \log_2(2)$$
$$= R_m$$

$$D = 8000 \text{ bps.}$$

2 bits  $\begin{matrix} 0 \\ 1 \end{matrix}$

1 valeur logique  
est codée par 1 valeur  
physique  $0 \rightarrow -V$   
 $1 \rightarrow +V$

+ Cas Manchester:

$$D = \frac{R_m}{k} \log_2(v)$$

or  $v = 2$ ,  $k = 2$ .

$$\Rightarrow D = \frac{R_m}{2} \log_2(2)$$

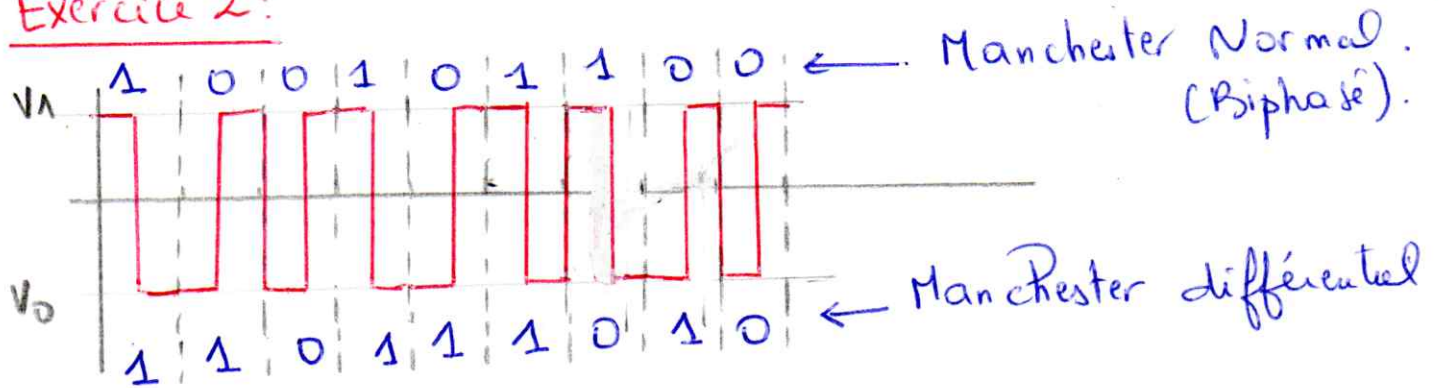
$$D = \frac{8000}{2} = 4000 \text{ bps.}$$

2 bits  
 $0 \wedge 1$

1 valeur logique  
est codée par 2 valeur  
physique exp:  $0 \rightarrow \begin{matrix} -V \\ +V \end{matrix}$

ou  $1 \rightarrow \begin{matrix} +V \\ -V \end{matrix}$

## Exercice 2:



## Exercice 4:

1)  $D = 20 \text{ Kbps}$ .

Codage manchester  $\Rightarrow k=2; v=2$ .

$$D = \frac{R_m}{k} \log_2(v) \Rightarrow R_m = \frac{D \times k}{\log_2(v)}$$

$$\text{AN: } R_m = \frac{20 \cdot 10^3 \times 2}{\log_2(2)}$$

"1"

$$R_m = 40 \cdot 10^3 \text{ bauds.}$$

2) Bande Passante =  $[10 \text{ KHz}, 22 \text{ KHz}]$ .

$$\Rightarrow W = 22 \cdot 10^3 - 10 \cdot 10^3 = 12 \text{ KHz}$$

b) Selon Nyquist on a :  $R_{\max} = 2 \cdot W$ .

$$\text{AN: } R_{\max} = 2 \times 12 \cdot 10^3$$

$$R_{\max} = 24 \cdot 10^3 \text{ Bauds.}$$

$$R_{\max} = 24 \cdot 10^3 \text{ Bauds} < R_{\min} = 40 \cdot 10^3 \text{ Bauds}$$

$\Rightarrow$  Le Transfert est impossible.

c) D'après Nyquist, on a:

$$C = W \log_2 \left( 1 + \frac{P_s}{P_B} \right).$$

$$\Rightarrow \frac{P_s}{P_B} = 2^{\frac{C}{W}} - 1.$$

$$\Rightarrow S/B = 10 \times \log_{10} \left( \frac{P_s}{P_B} \right).$$

$$S/B = 10 \times \log_{10} \left( 2^{\frac{C}{W}} - 1 \right).$$

$$\text{AN: } S/B = 10 \times \log_{10} \left( 2^{\frac{20 \cdot 10^3}{12 \cdot 10^3}} - 1 \right).$$

$$S/B = 10 \log_{10} \left( 2^{\frac{20}{12}} - 1 \right).$$

$$S/B = 3.37 \text{ dB.}$$

### Exercice 7:

1)  $R_m = 1200$  bauds.  $\Rightarrow$  l'inconnu est  $v$  ?  
 $D = 2400$  bps.  
 $k = 1$

$$\text{on } D = \frac{R_m}{k} \log_2(v)$$

$$\Rightarrow \log_2(v) = \frac{k \cdot D}{R_m}.$$

$$v = 2^{\frac{k \cdot D}{R_m}}.$$

$$\text{AN: } v = 2^{\frac{1 \times 2400}{1200}} = 4.$$

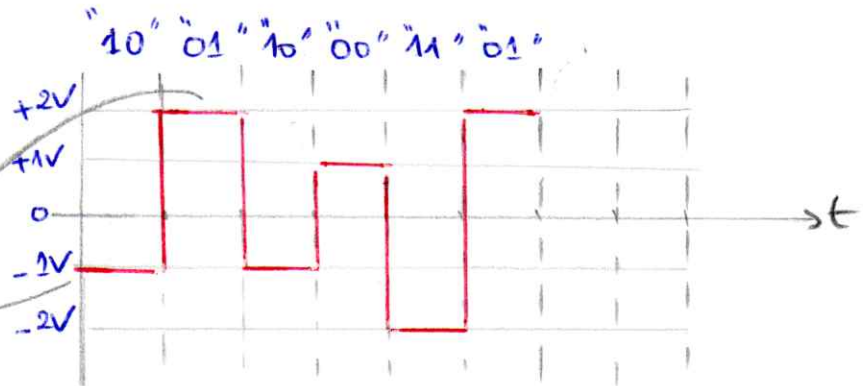
$v = 4 \Rightarrow 4$  états possible  $\Rightarrow$  nombre de bit par symbol = 2 bits.



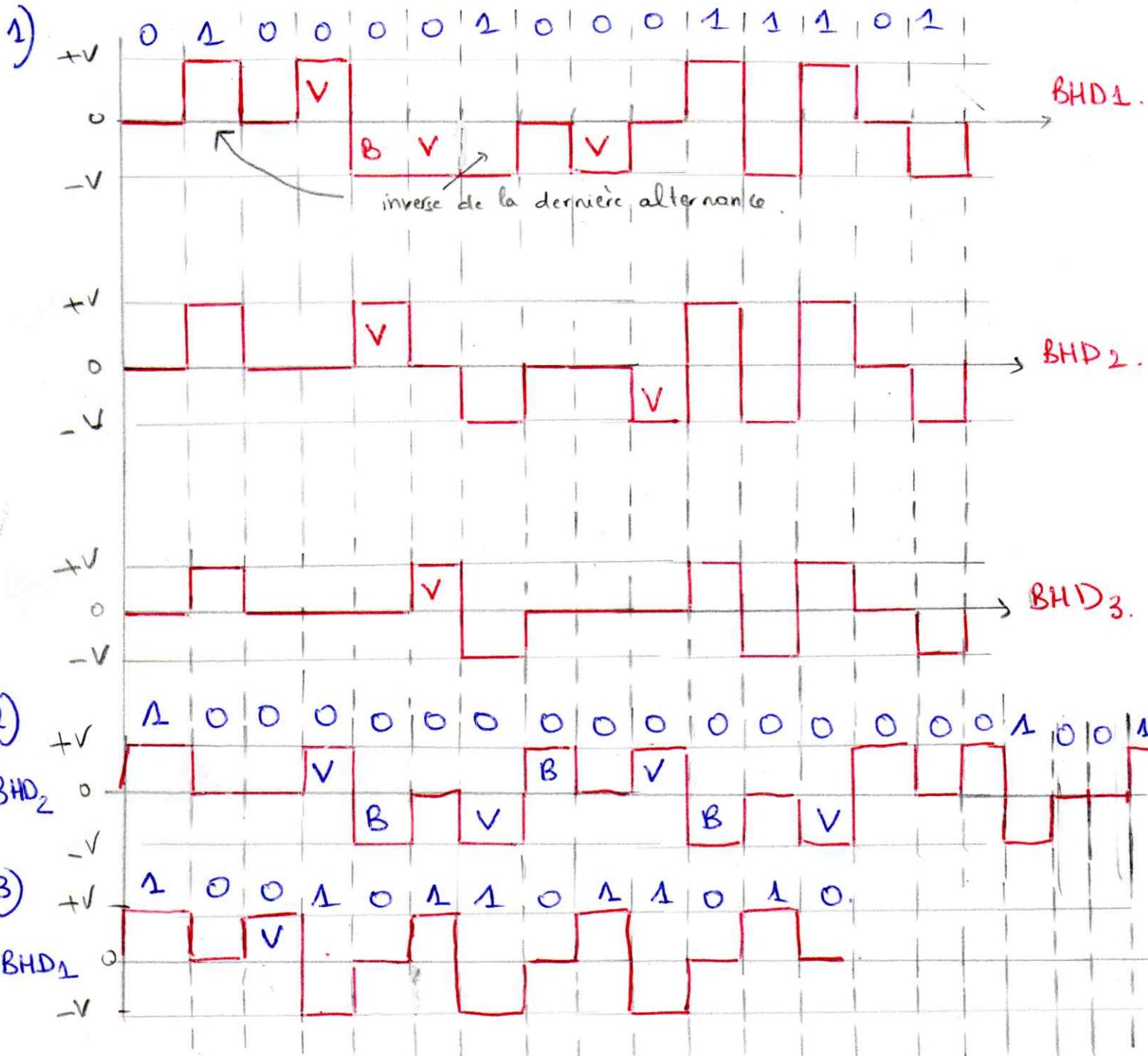
⇒ il s'agit d'un codage par bloc NRZ avec 2 bits par symbol.

Etats possible :

00 → +1V  
01 → +2V  
10 → -1V  
11 → -2V



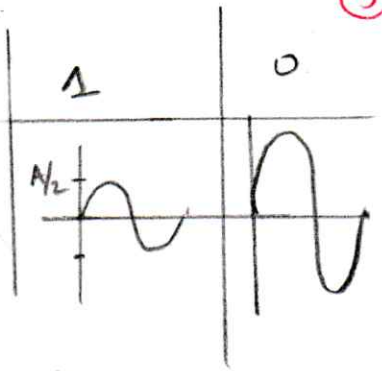
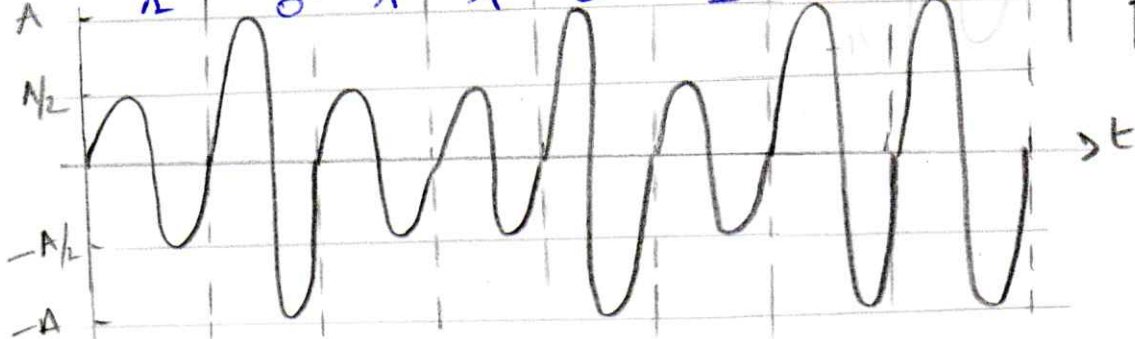
### Exercice 5 :



## Exercice 6:

### \* Modulation d'amplitude

Bit 0  $\rightarrow A$ ; Bit 1  $\rightarrow A/2$ .

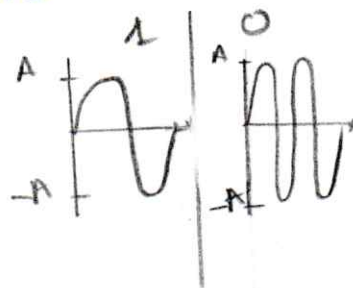
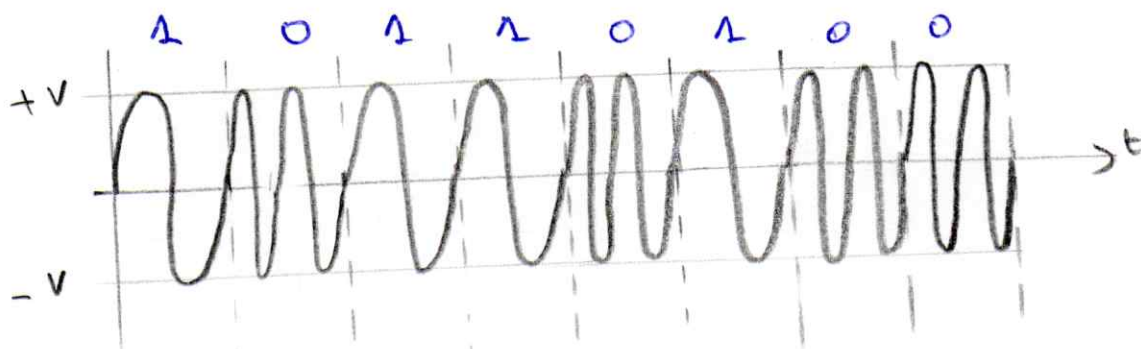


### \* Modulation de fréquence

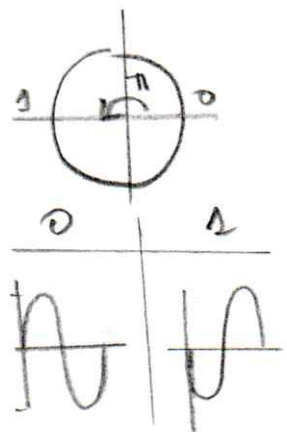
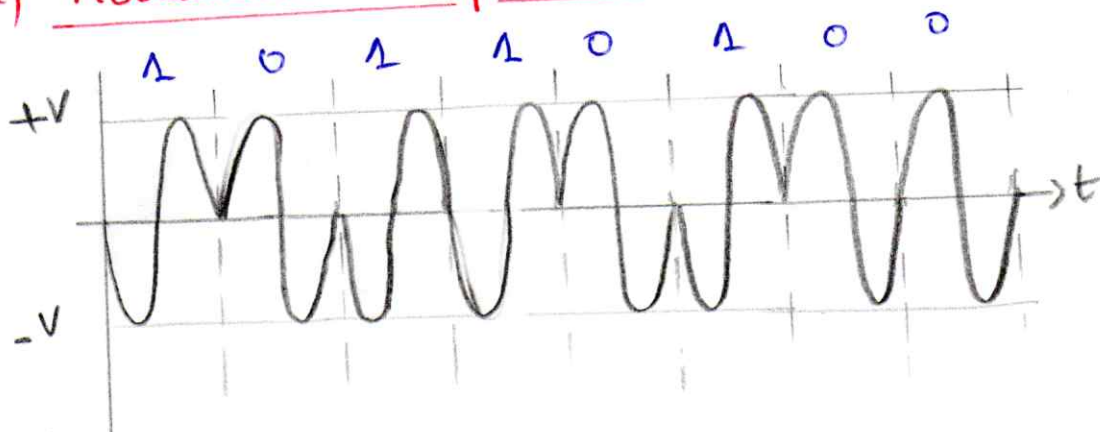
Bit 0  $\rightarrow f$ ; Bit 1  $\rightarrow f/2$ .

$$f_0 = f \Rightarrow f_0 = 2f_1 \Rightarrow \frac{1}{T_0} = \frac{2}{T_1} \Rightarrow T_1 = 2T_0$$

$$f_1 = \frac{f}{2} \Rightarrow 2 \times f_1 = f_0$$



### \* Modulation de phase



## Exercice 8

- Modulation Mixte phase + Amplitude

- 8 états de phases + 2 Amplitude  $\Rightarrow$  3 bits par symbole.

